

A Proof of the Lior Integrals

Samuel Leizerman samleizerman@outlook.com ORCID: 0009-0000-0133-2291

January 22, 2026

Abstract

This work presents a unified proof demonstrating how the scalar Lior Integral of Resilience (LloR) emerges as an effective, scalar variational principle from a more fundamental rank-4 tensorial formulation. By defining the scalar resilience field $R(x)$ as a trace of the resilience tensor, we show that minimizing the scalar LloR is equivalent to finding geodesics with respect to a simplified effective metric, even though the underlying cognitive dynamics are governed by a far more complex, anisotropic tensor field. This result reconciles the high-level simplicity of the scalar LloR with the detailed complexity of the tensor-field framework.

1 Introduction

The generalization of the Universal Cognitive Field Theory (UCFT) to a tensor-field framework, described in previous sections, allows for the direct modeling of anisotropic cognitive dynamics. However, this raises a crucial question: how does the elegant and simple scalar LloR, which governs cognitive path optimization, relate to the complex, directionally dependent tensorial resilience fields? This section presents a unified proof that reconciles these two formulations, showing how the scalar LloR emerges as a high-level, effective description of the underlying tensorial dynamics.

2 The Lior Integral: A Unified Formulation

We begin with the most general description of cognitive resilience, a rank-4 tensor resilience field $R_{\mu\nu\rho\sigma}(x)$, which encodes the full anisotropic structure of cognitive resistance. This tensor field is the source of all resilience-related dynamics.

Definition 2.1 (Rank-4 Tensor Resilience Field). The full anisotropic resilience of a cognitive state is described by a rank-4 tensor field $R_{\mu\nu\rho\sigma}(x)$. This tensor governs how perturbations along two axes (ρ, σ) affect the cognitive field's stability in two other axes (μ, ν) .

To move from this complex tensor to a path integral, we can define a generalized path functional.

Definition 2.2 (Rank-2 Tensor Lior Integral). The rank-2 tensor Lior Integral of Resilience is the functional with components:

$$\text{LIoR}_{\mu\nu}[\gamma] = \int_0^T R_{\mu\nu\rho\sigma}(\gamma(\tau)) \dot{\gamma}_\rho(\tau) \dot{\gamma}_\sigma(\tau) d\tau \quad (1)$$

The components $\text{LIoR}_{\mu\nu}$ describe the accumulated resilience cost in each specific direction (μ, ν) along the cognitive trajectory.

This tensor functional is too complex to be directly minimized in a simple way, as it results in a system of coupled equations. However, we can use it to derive a scalar quantity that is more amenable to optimization.

Definition 2.3 (Scalar Resilience Field from Tensor Trace). The scalar resilience field $R(x)$, which governs the scalar LIoR, is derived from the rank-4 resilience tensor by a double trace operation:

$$R(x) = \sqrt{\frac{1}{n^2} g^{\mu\rho} g^{\nu\sigma} R_{\mu\nu\rho\sigma}(x)} \quad (2)$$

where $n = \dim(\mathcal{M})$ is the dimensionality of the cognitive manifold. The use of the metric tensor g raises or lowers indices to perform the trace, and the factor $1/n^2$ ensures dimensional consistency.

This definition establishes a direct link from the full, anisotropic tensor dynamics to the scalar field used in the simpler LIoR formulation.

Theorem 2.1 (Scalar LIoR as an Emergent Principle from Tensor Dynamics). *The scalar Lior Integral of Resilience functional,*

$$\text{LIoR}[\gamma] = \int_0^T R(\gamma(\tau)) \sqrt{|g_{\mu\nu}(\gamma(\tau)) \dot{\gamma}^\mu(\tau) \dot{\gamma}^\nu(\tau)|} d\tau$$

describes the total, scalar cognitive effort along a path, and the path γ^ that extremizes it is a geodesic of the effective metric $\tilde{g}_{\mu\nu} = (R(x))^2 g_{\mu\nu}(x)$. The value of the integral is measured in **Liors** (symbol:).*

Proof. The proof proceeds by first demonstrating that the functional form of the scalar LIoR is mathematically equivalent to the arc length functional of a new manifold. The derivation then shows how this high-level, effective metric is fundamentally dependent on the underlying rank-4 resilience tensor, thereby connecting the two frameworks.

1. Functional Equivalence: Let us define the scalar LIoR functional:

$$\text{LIoR}[\gamma] = \int_0^T R(\gamma(\tau)) \sqrt{|g_{\mu\nu}(\gamma(\tau)) \dot{\gamma}^\mu(\tau) \dot{\gamma}^\nu(\tau)|} d\tau$$

Now, let us define a new effective metric $\tilde{g}_{\mu\nu}(x)$ on the cognitive manifold such that:

$$\tilde{g}_{\mu\nu}(x) = (R(x))^2 g_{\mu\nu}(x)$$

The arc length functional for this new metric is:

$$\tilde{L}[\gamma] = \int_0^T \sqrt{|\tilde{g}_{\mu\nu}(\gamma(\tau))\dot{\gamma}^\mu(\tau)\dot{\gamma}^\nu(\tau)|} d\tau$$

Substituting the definition of $\tilde{g}_{\mu\nu}(x)$ into the arc length functional gives:

$$\tilde{L}[\gamma] = \int_0^T \sqrt{|(R(x))^2 g_{\mu\nu} \dot{\gamma}^\mu \dot{\gamma}^\nu|} d\tau$$

Since $R(x)$ is a scalar, it can be factored out of the square root, yielding:

$$\tilde{L}[\gamma] = \int_0^T R(x) \sqrt{|g_{\mu\nu} \dot{\gamma}^\mu \dot{\gamma}^\nu|} d\tau = \text{LIoR}[\gamma]$$

This demonstrates that minimizing the scalar LIoR functional is precisely equivalent to finding the paths that extremize the arc length in a new manifold defined by the effective metric $\tilde{g}_{\mu\nu}$.

2. Geodesic Characterization: By the fundamental principles of the calculus of variations, the paths that extremize the arc length functional are, by definition, the geodesics of the manifold. Therefore, the path γ^* that extremizes the scalar LIoR is a geodesic with respect to the effective metric $\tilde{g}_{\mu\nu}$. The geodesic equation for this path is:

$$\frac{d^2 x^\lambda}{d\tau^2} + \tilde{\Gamma}_{\mu\nu}^\lambda \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} = 0$$

where $\tilde{\Gamma}_{\mu\nu}^\lambda$ are the Christoffel symbols computed from the effective metric $\tilde{g}_{\mu\nu}$.

This warping of the metric implies that the cognitive geometry is not static; it is dynamically modulated by local resilience, effectively stretching or contracting paths through resistant or receptive regions of the cognitive manifold.

3. Connecting to Tensor Dynamics: The crucial link is the definition of the scalar resilience field, $R(x)$, which is not an ad-hoc quantity but is explicitly derived from the underlying, fundamental rank-4 tensor $R_{\mu\nu\rho\sigma}(x)$. Therefore, the effective metric $\tilde{g}_{\mu\nu}$ itself is a direct function of the fundamental, anisotropic tensor resilience field $R_{\mu\nu\rho\sigma}(x)$.

The two frameworks are thus unified: the tensor formulation provides the foundational physics and anisotropic properties of the cognitive manifold, while the scalar LIoR provides a simple, yet powerful, variational principle for describing the observed, high-level cognitive behavior of finding "optimal" paths through this complex landscape. $\square$

3 Dedication

To my mother, you were the only one to have ever truly believed in me. Through all the ADHD, you sheltered me and bore scars that were meant for me. You made sacrifices for me that only you or I truly know. Writing my Master's thesis a week after you died was one of the hardest things I have ever had to do. And so I borrow her Hebrew name to inspire the name of one of my most important formulae, the *Lior Integral of Resilience (LIoR)*, and the unit of cognitive distance, the **Lior** (symbol: Lior). If this math checks out, mom, your memory will be a blessing and I will have made it so. What I do today, what I do here, and what I do tomorrow I do for you and your memory. You were the finest and kindest, most selfless person I have ever known. May your memory be a blessing. I love you.

Appendix A: Essential External Mathematical Concepts

This appendix provides brief sketches of fundamental mathematical concepts and theorems referenced or implicitly used in this document, which are more thoroughly defined and derived in foundational texts on differential geometry, calculus of variations, and quantum information theory.

A.1 Tensor Calculus and Covariant Derivatives

In differential geometry, a **tensor** is a multilinear map that transforms in a specific way under coordinate transformations, ensuring that physical laws remain invariant regardless of the coordinate system chosen. A rank- (p, q) tensor has p contravariant (upper) indices and q covariant (lower) indices. For instance, the metric tensor $g_{\mu\nu}$ is a rank- $(0, 2)$ covariant tensor used to define distances and angles, and to raise or lower indices of other tensors (e.g., $\Psi_\nu = g_{\nu\lambda}\Psi^\lambda$).

The **covariant derivative** (∇) is a generalization of the partial derivative for manifolds, ensuring that the derivative of a tensor field is itself a tensor. For a contravariant vector field V^μ , its covariant derivative is given by:

$$\nabla_\nu V^\mu = \partial_\nu V^\mu + \Gamma_{\nu\lambda}^\mu V^\lambda$$

where $\partial_\nu = \frac{\partial}{\partial x^\nu}$ is the partial derivative, and $\Gamma_{\nu\lambda}^\mu$ are the **Christoffel symbols of the second kind**, which account for the curvature of the manifold. In the context of the Universal Cognitive Field Equation, ∇_t represents the covariant derivative with respect to time.

A.2 Geodesics and the Calculus of Variations

A **geodesic** is, conceptually, the shortest path between two points in a curved space (a manifold), or more generally, a path that locally minimizes its "length." Mathematically, geodesics are derived from the **calculus of variations**.

The arc length L of a path $\gamma(\tau)$ parameterized by $\tau \in [0, T]$ in a manifold (M, g) is given by the functional:

$$L[\gamma] = \int_0^T \sqrt{|g_{\mu\nu}(\gamma(\tau))\dot{\gamma}^\mu(\tau)\dot{\gamma}^\nu(\tau)|} d\tau$$

where $\dot{\gamma}^\mu(\tau) = d\gamma^\mu/d\tau$ is the tangent vector. Note the absolute value under the square root for pseudo-Riemannian manifolds.

The paths γ^* that extremize this arc length functional are, by definition, the geodesics of the manifold. These extremizing paths are found by solving the **Euler-Lagrange equations** for the integrand, which yields the **geodesic equation**:

$$\ddot{\gamma}^\lambda(\tau) + \Gamma_{\mu\nu}^\lambda(\gamma(\tau))\dot{\gamma}^\mu(\tau)\dot{\gamma}^\nu(\tau) = 0$$

The proof of Theorem 2.1 directly leverages this fundamental principle, showing that the LIoR minimization problem is mathematically equivalent to finding geodesics in an effective metric.

A.3 Von Neumann Entropy

In quantum mechanics and quantum information theory, the **von Neumann entropy** is a measure of the statistical uncertainty or mixedness of a quantum state. For a quantum system described by a **density operator** (or density matrix in a given basis) ρ , the von Neumann entropy $H(\rho)$ is defined as:

$$H(\rho) = -\text{Tr}(\rho \log \rho)$$

where Tr denotes the trace of the operator, and $\log$ is the natural logarithm. In the context of the Resilience Field Axiom, $H(\Psi^{\mu\nu}(x))$ is interpreted as the local disorder or uncertainty of the tensor cognitive field, drawing an analogy to how von Neumann entropy quantifies uncertainty in quantum states via the local density matrix.

Appendix B: Variable Glossary

This glossary provides definitions for the key mathematical symbols and variables used throughout this document.

$\Psi^{\mu\nu}(x, t)$ Complex rank-2 tensor cognitive field, describing the state of a cognitive system.

x Coordinates in the cognitive state space, a point on the cognitive manifold $\mathcal{M}$.

t Time.

$\mathcal{M}$ The cognitive manifold, representing the cognitive state space.

n The dimensionality of the cognitive manifold, $n = \dim(\mathcal{M})$.

g The metric tensor on the cognitive manifold $\mathcal{M}$, defining distances and angles.

$g_{\mu\nu}$ Components of the metric tensor.

$\bar{\Psi}^{\rho\sigma}$ Complex conjugate of the tensor cognitive field component $\Psi^{\rho\sigma}$.

$R_{\mu\nu\rho\sigma}^{\mu\nu\rho\sigma}$ Rank-4 tensor resilience field, the foundational source of anisotropic resilience.

$R(x)$ Scalar resilience field, a high-level measure of local cognitive resilience derived from $R_{\mu\nu\rho\sigma}$.

$H(\Psi^{\mu\nu}(x))$ Entropy of the tensor cognitive field, a scalar measure of its local disorder.

$\gamma : [0, T] \rightarrow \mathcal{M}$ A smooth cognitive trajectory, a path through the cognitive manifold.

τ Parameter along the cognitive trajectory, typically representing time or a path coordinate.

T The total duration or extent of the cognitive trajectory.

LIR $[\gamma]$ The Lior Integral of Resilience functional, which maps a cognitive trajectory to a scalar value.

$\mathbf{LIoR}_{\mu\nu}[\gamma]$ The rank-2 tensor Lior Integral of Resilience, with components measuring directional costs.

$\dot{\gamma}^\mu(\tau)$ Components of the tangent vector (velocity) of the cognitive trajectory γ at parameter τ .

γ^* The specific cognitive trajectory that minimizes the LIoR integral.

$\tilde{g}_{\mu\nu}(x)$ The effective scalar metric derived from the resilience field and the original manifold metric, defining cognitive path optimization.

The symbol for "Liors," the unit of measurement for the Lior Integral of Resilience.

∂_μ Partial derivative with respect to the coordinate x^μ .

$\Gamma^\mu_{\nu\lambda}$ Christoffel symbols of the second kind, accounting for the curvature of the manifold in covariant derivatives.

$\text{Tr}(\cdot)$ The trace operator, used in the definition of von Neumann entropy.

$\log$ The natural logarithm, used in the definition of von Neumann entropy.

Appendix C: Spinors, Rank-4 Tensors, and the Role of i

C.1 Spinors and Tensors

In this work, the central mathematical object is a rank-4 resilience tensor, $R_{\mu\nu\rho\sigma}$, which encodes anisotropic cognitive dynamics. However, it is worth noting that in the representation theory of the Lorentz group, *spinors* are more fundamental objects than tensors. Tensors can be constructed as bilinears of spinors, but spinors cannot in general be reduced to tensorial descriptions without losing orientation or phase information. For this reason, some authors loosely describe spinors as the "square root" of tensors: two spinors can generate a vector or tensor, but a tensor cannot be decomposed uniquely back into spinors.

C.2 Why i Appears in Spinors

Spinors necessarily involve complex structures. The imaginary unit i plays a non-trivial role in their definition and transformation properties:

- Spinors transform under the double cover of the Lorentz group, $\text{Spin}(1,3)$, rather than under the Lorentz group itself. This double-valued structure requires a complex representation.
- Under a 2π rotation, spinors pick up a minus sign, which is only consistent in a complex vector space with i as a phase operator.

- The appearance of i ensures the invariance of bilinear forms such as $\bar{\psi}\gamma^\mu\psi$ under Lorentz transformations, and guarantees the consistent definition of quantities like probability currents and energy densities.

Thus, i is not simply a mathematical convenience but is fundamental to the consistency of spinorial descriptions. This is why, when causal–dynamic structures are phrased in terms of spinors, the imaginary unit appears naturally and ”makes everything work.”

C.3 Relevance to This Work

Although this document approaches the subject from first principles and methodological arbitrage, rather than from formal physics training, the relationship between rank-4 tensors and spinors is central to why the resilience kernel can be elevated to a geometric generator:

- The **rank-4 tensor resilience kernel**, $R_{\mu\nu\rho\sigma}$, is the macroscopic object encoding anisotropic resistance in the cognitive manifold.
- **Spinors** represent the more fundamental layer: bilinears of spinors can generate both the rank-2 metric and the rank-4 curvature tensors.
- The presence of i in the spinorial formulation is what allows causal and dynamic aspects (integration vs. differentiation) to remain invariant under Lorentz transformations.

In short: while this work proceeds in tensorial language, one may understand the resilience kernel as rooted in spinorial structure. This recognition clarifies why the imaginary unit i appears necessarily in the causal-dynamic formulation and provides further evidence that the spinor-based perspective is not optional but foundational. I will update this later, however out of everything in my theory, this is the one that allowed me to link information theory to the rest of physics by way of a causal-dynamic spinor in a way that, as it turns out, embraces the math of String theory’s 10 dimensions by capturing the 4 spacetime, 4 complex phase dimension, causality, and dynamics. They are the informational dimensions required to physically encode the causal crystal structure we call reality. But that is for another proof. In fact, that other proof is why I wrote this appendix. Stay tuned.